

Pharmaceutical Reform and Pharmacy Entry: Evidence from Hospitals in China

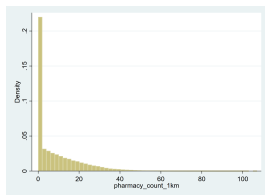
Wenxia Chen

April 15, 2026

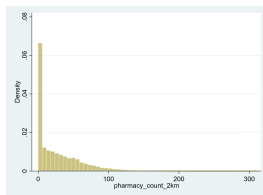
Distribution of Policy Adoption Years

Policy Year	Frequency	Percent (%)	Cumulative (%)
Never treated (0)	86,027	34.37	34.37
2012	6,494	2.59	36.97
2013	1,620	0.65	37.62
2014	34,471	13.77	51.39
2015	43,131	17.23	68.62
2016	16,619	6.64	75.26
2017	61,904	24.74	100.00
Total	250,266	100.00	

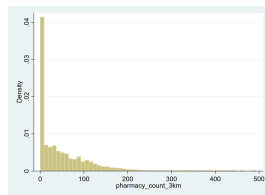
Distribution of Pharmacy Density Around Hospitals



(a) 1km



(b) 2km



(c) 3km

Note: Distribution is highly right-skewed with mass at zero.

Empirical Strategy: TWFE

$$Y_{ict} = \beta_0 + \beta_1 \text{Reform}_{ct} + \delta_i + \delta_t + \epsilon_{ict}$$

- ▶ Y_{ict} : Number of pharmacies around hospital i in county c at time t
- ▶ Reform_{ct} : Policy exposure at the county-year level
- ▶ δ_i : hospital fixed effects
- ▶ δ_t : Year fixed effects

Traditional TWFE may be biased under staggered adoption

Empirical Strategy: CS-DID

$$ATT(g, t) = \mathbb{E}[Y_t(1) - Y_t(0) \mid G = g]$$

- ▶ g : First year of reform (cohort)
- ▶ t : Year
- ▶ Compare treated units with not-yet-treated units
- ▶ Allows for heterogeneous treatment effects
- ▶ Avoids bias from two-way fixed effects under staggered adoption

Estimate dynamic and cohort-specific treatment effects

Baseline DID Results:TWFE

OLS

- ▶ Outcome: Pharmacy entry around hospitals
- ▶ Sample: score=10 | score=8
- ▶ Fixed effects: hospital and year
- ▶ Standard errors : clustered at hospital level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
did	-0.433*** (0.103)	-0.942*** (0.263)	-1.101* (0.449)
N	193010	193010	193010
R-sq	0.884	0.895	0.903

Baseline DID Results:TWFE

OLS

- ▶ Outcome: Pharmacy entry around hospitals
- ▶ Sample: score=10 | score=8
- ▶ Fixed effects: Hospital and year
- ▶ Standard errors : clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
did	-0.433** (0.161)	-0.947* (0.441)	-1.113 (0.776)
N	192868	192868	192868
R-sq	0.884	0.895	0.903

Baseline DID Results: CS-DID

Callaway and Sant'Anna(2021)

- ▶ Outcome model : least squares
- ▶ Control: never treated
- ▶ Treatment model: inverse probability
- ▶ Standard errors : clustered at hospital level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
ATT	-0.0736 (0.0985)	-.320526 (.2412168)	-.5840457 (.4028106)
N	124137	124137	124137

Baseline DID Results: CS-DID

Callaway and Sant'Anna(2021)

- ▶ Outcome model : least squares
- ▶ Control: never treated
- ▶ Treatment model: inverse probability
- ▶ Standard errors : clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
ATT	-.0735517 (.0985356)	-.320526 (.2412168)	-.5840457 (.4028106)
N	124137	124137	124137

Baseline DID Results: PPML

PPML

- ▶ Outcome: Pharmacy entry around hospitals
- ▶ Fixed effects: hospital and year
- ▶ Sample: score=10 | score=8
- ▶ Standard errors : clustered at hospital level

	(1)	(2)	(3)
	pharmacy~1km	pharmacy~2km	pharmacy~3km
did	0.042*** (0.009)	0.049*** (0.009)	0.058*** (0.009)
N	173450	181287	183995

Baseline DID Results: PPML

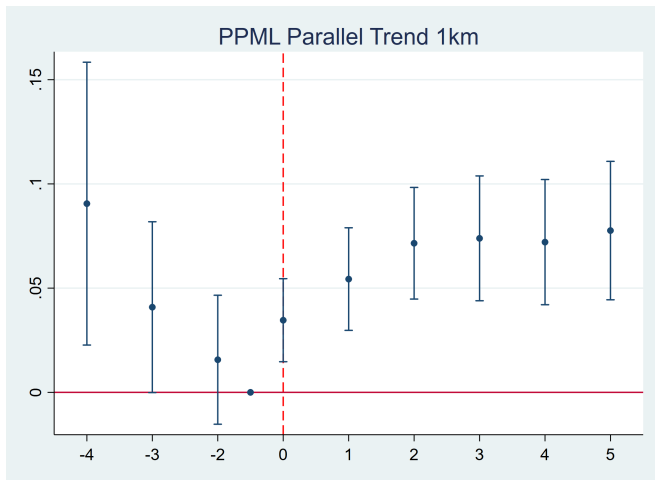
PPML

- ▶ Outcome: Pharmacy entry around hospitals
- ▶ Fixed effects: hospital and year
- ▶ Sample: score=10 | score=8
- ▶ Standard errors : clustered at county level

	(1)	(2)	(3)
	pharmacy~1km	pharmacy~2km	pharmacy~3km
did	0.043*** (0.016)	0.050*** (0.015)	0.058*** (0.016)
N	173319	181145	183853

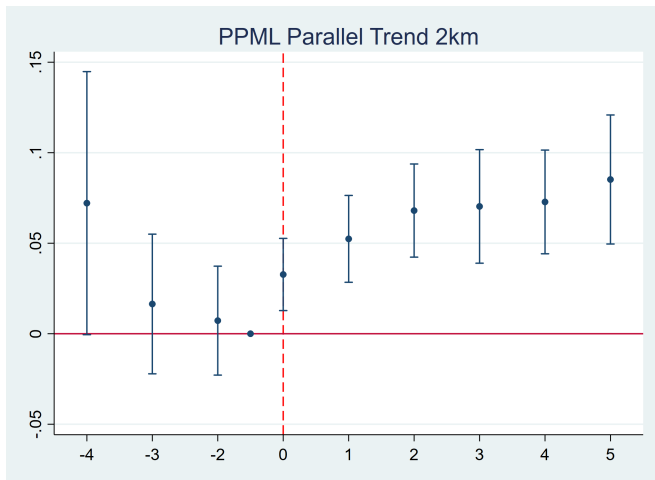
Event Study: PPML

- Outcome: Pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



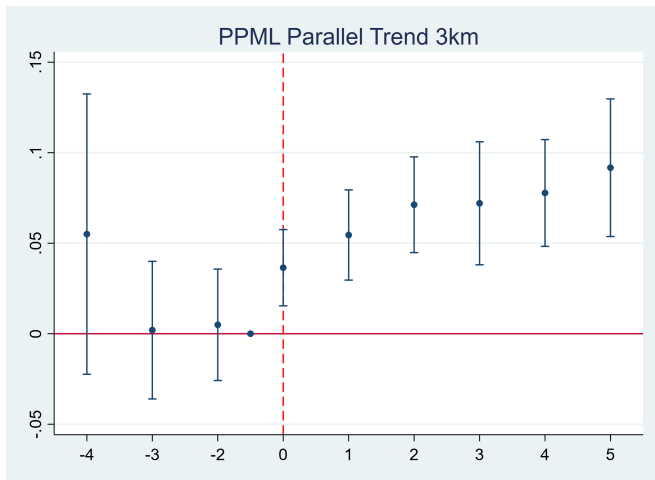
Event Study: PPML

- ▶ Outcome: Pharmacy entry around hospitals (within 2km)
- ▶ Fixed effects: hospital and year
- ▶ Standard errors : clustered at county level



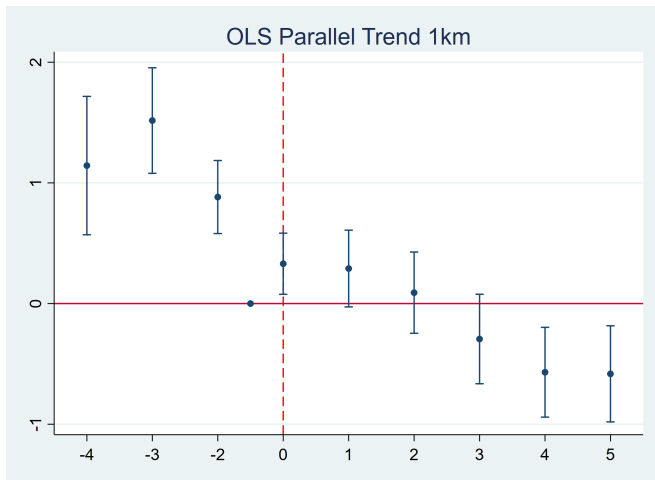
Event Study: PPML

- ▶ Outcome: Pharmacy entry around hospitals (within 3km)
- ▶ Fixed effects: hospital and year
- ▶ Standard errors : clustered at county level



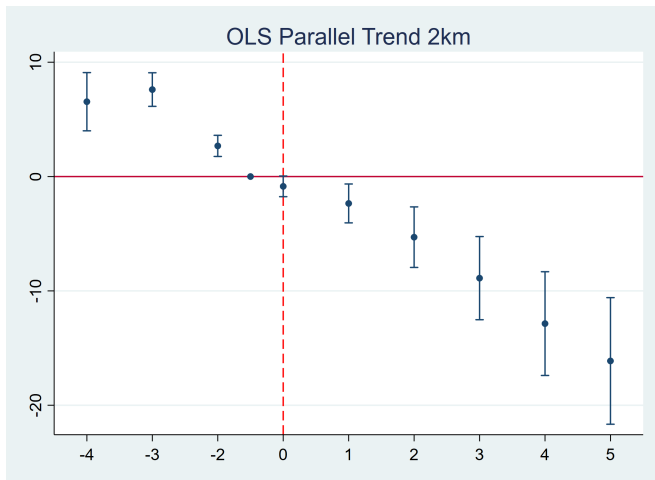
Event Study: TWFE

- Outcome: Pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



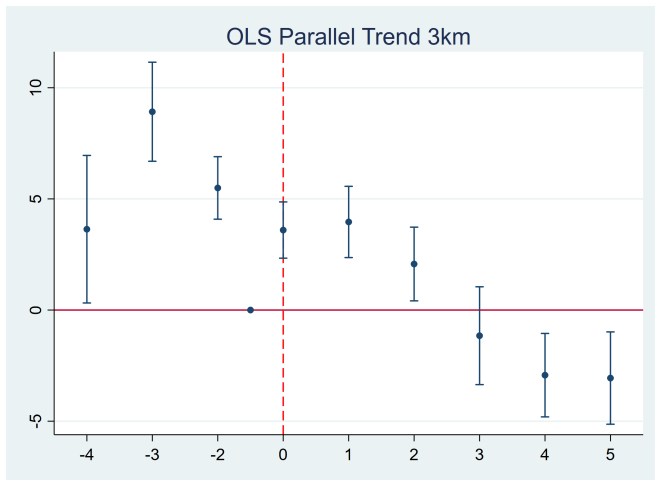
Event Study: TWFE

- Outcome: Pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



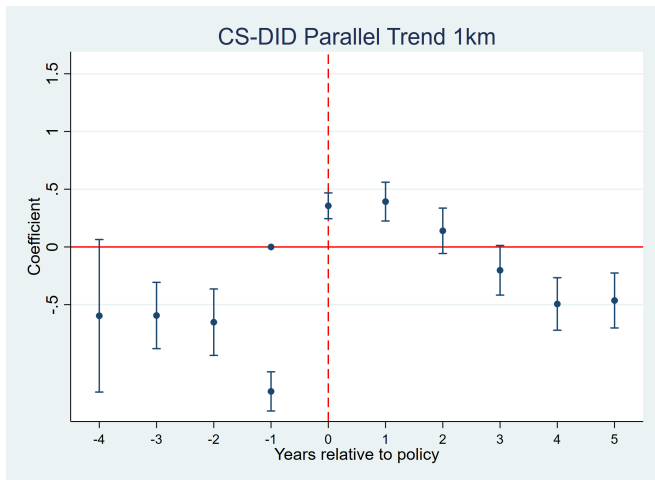
Event Study: TWFE

- ▶ Outcome: Pharmacy entry around hospitals (within 3km)
- ▶ Fixed effects: hospital and year
- ▶ Standard errors : clustered at county level



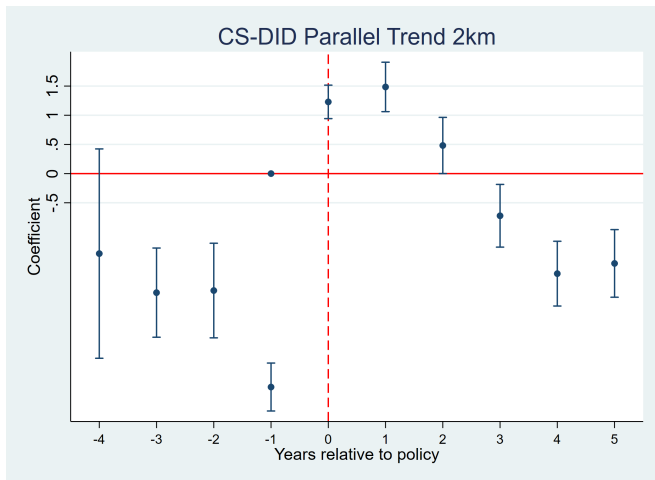
Event Study:CS

- ▶ Outcome: Pharmacy entry around hospitals (within 1km)
- ▶ Fixed effects: hospital and year
- ▶ Standard errors : clustered at county level



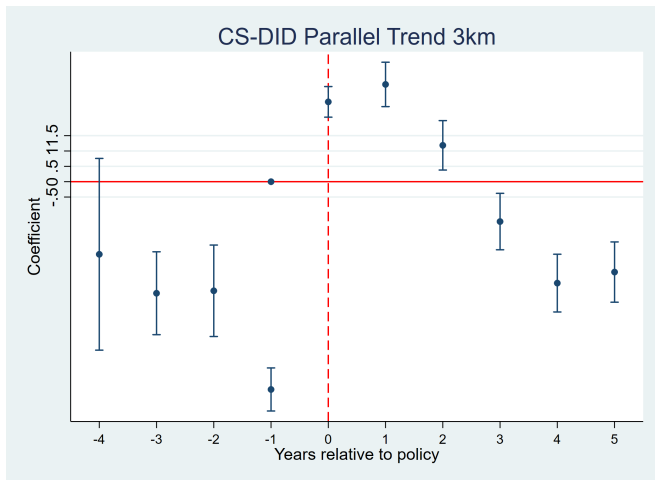
Event Study:CS

- Outcome: Pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



Event Study:CS

- Outcome: Pharmacy entry around hospitals (within 3km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



Mechanism: Chain Pharmacies(OLS)

- ▶ Outcome: Chain pharmacies entry around hospitals
- ▶ Fixed effects: hospital and year
- ▶ Standard errors: clustered at county level

	(1) Chain pharmacy ~1km	(2) Chain pharmacy ~2km	(3) Chain pharmacy ~3km
did	0.012 (0.088)	0.382 (0.261)	1.017* (0.479)
N	188,410	188,410	188,410
R-sq	0.682	0.719	0.733

Mechanism: Non-Chain Pharmacies

- ▶ Outcome: Non-chain pharmacies entry around hospitals
- ▶ Fixed effects: hospital and year
- ▶ Standard errors: clustered at county level

	(1) Non-chain pharmacy ~1km	(2) ~2km	(3) ~3km
did	-0.537*** (0.117)	-1.553*** (0.310)	-2.495*** (0.538)
N	188,410	188,410	188,410
R-sq	0.828	0.833	0.836

Mechanism: Chain Pharmacies (Share)

- ▶ Outcome: Share of chain pharmacies around hospitals
- ▶ Fixed effects: hospital and year
- ▶ Standard errors: clustered at county level

	(1) Chain share ~1km	(2) Chain share ~2km	(3) Chain share ~3km
did	0.019*** (0.004)	0.021*** (0.004)	0.022*** (0.004)
N	159,005	170,604	174,613
R-sq	0.788	0.840	0.851

Mechanism: Chain Pharmacies (PPML)

- ▶ Outcome: Chain pharmacies around hospitals (count)
- ▶ Estimator: PPML with hospital and year fixed effects
- ▶ Standard errors: clustered at county level

	(1)	(2)	(3)
	Chain pharmacy ~1km	Chain pharmacy ~2km	Chain pharmacy ~3km
did	0.334*** (0.088)	0.353*** (0.082)	0.358*** (0.079)
N	139,926	149,752	153,318

Mechanism: Non-chain Pharmacies (PPML)

- ▶ Outcome: Non-chain pharmacies around hospitals (count)
- ▶ Estimator: PPML with hospital and year fixed effects
- ▶ Standard errors: clustered at county level

	(1)	(2)	(3)
	Non-chain ~1km	Non-chain ~2km	Non-chain ~3km
did	0.007 (0.013)	0.010 (0.013)	0.016 (0.013)
N	171,807	180,668	183,617

Mechanism: Chain Pharmacies (Share, PPML)

- ▶ Outcome: Share of chain pharmacies around hospitals
- ▶ Estimator: PPML with hospital and year fixed effects
- ▶ Standard errors: clustered at county level

	(1)	(2)	(3)
	Chain share ~1km	Chain share ~2km	Chain share ~3km
did	0.205*** (0.061)	0.229*** (0.057)	0.242*** (0.057)
N	129,869	141,824	146,097

Summary

- ▶ DID: OLS negative and significant ,CS negative but insignificant ,PPML positive and significant
- ▶ Event Study: PPML, supporting causal inference
- ▶ How to adjust event study

The reform increases pharmacy entry around hospitals